

PART A
(PART A: TO BE REFERRED BY STUDENTS)

Practical No. 5

A.1 Aim:

1. To understand Software Quality assurance activities of review and test scenario and test case preparation.

A.2 Prerequisite:

1. Problem statement mentioned in Practical No. 1

A.3 Tasks:

1. For the problem statement allocated in Practical No. 1, undergo the review activity of requirement, design, test scenario preparation and test case preparation.

A.3 Outcome:

1. Students understand the different software quality assurance activities namely review, test scenario and test case preparation.

A.4 Theory:

Peer Review

1. Refer to PD-07-Peer_Review.docx process for peer review.
2. Refer to PD-04-Testing.docx process document for testing.
3. PD-07-01-Review_Log.xlsx - peer review log.

Test Case Preparation

1. PD-04-Testing.docx testing process understanding
2. PD-04-01-Testing_Guidelines.docx – Testing guidelines document
3. PD-04-03-Test_Team_Criteria_Guidelines.docx – Criteria for the independent test team
4. PD-04-04-Testing_Checklist.xlsx – Testing checklist
5. PD-04-02-Test_Cases_And_BugReport.xlsx test cases, test scenario and bug report preparation template
6. PD-04-05-Test_case_Matrix.xlsx – Matrix approach of test case preparation.

PART B

(PART B : TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed (chandravadanp@outlook.com) to the concerned lab in charge faculties at the end of the practical in case there is no Black board access available)

Roll. No.	Name:
Class	Batch:
Date of Experiment:	Date of Submission:
Grade:	

B.1 Excel sheet developed by a student:

SOFTWARE QUALITY ASSURANCE ACTIVITIES	Yes	No	N/A
I. Risk Management			
Risk are recorded and prioritized	X		
The countermeasure to avoid risk must be defined	X		
II. Test Estimation			
Project tasks are break out into work product for estimation	X		
The estimation must be approved by Manager		X	
III. Test Planning			
Test test scope and Test objective are defined	X		
Test Planning are documented	X		
IV. Test Organization			
Resource for Testing are defined	X		
Project members and their roles in project are defined	X		
V. Project Monitoring			
The Test Manager monitor the Project by collecting metrics from members		X	
All the metrics of project must be collected and documented		X	
VI. Issues Management			
Any issues happened in the project must be reported to management board	X		
The countermeasure for the issues must be defined and approved	X		
VII. Test report			
The test report must be documented	X		
The result must be acknowledged and approved by Manager		X	
VIII. REVIEWS and AUDITS			
SQA reviews and audits are documented.		X	
Process SQA reviews occur.		X	
SQA reviews of the work product are performed.	X		
SQA activities are reported to the project manager.		X	

YES	No	N/A	SQA result
11	7	0	
61.11%			FAIL

Ver 2.01

Ver 2.02 / 25-Jul-04

Review Log

Project ID:	CALC-01	Project Name:	Java Calculator Application
Reviewable item:	Source Code (Calculator.java), UI Design	Author(s):	Anshuman Chakravorty
Reviewed by:	Peer Reviewer	Reviewed on:	26-Mar-26
Moderator:	Faculty	Accepted on:	27-Mar-26
Version of reviewable:	v1.0	No. of defects detected:	6
Reviewable LOC/Pages:	250 LOC	LOC/Pages to be reworked:	40
Review stage:	CO	Follow up action:	Redo and Re-review

Details of defects:

Sr. No.	Doc. Id/ Program Id	Location	Description, Optional for document reviews	Category, By Reviewer	Severity, By Reviewer	Cause, by Author	Introduced at	Status	Fixed On Date	Closed/ Transferred On, Date	Remarks
1	Calculator.java	Division Function	Division by zero not handled properly	Logic Error	High	Missing condition check	Coding	Fixed	26-03-2026	27-03-2026	Added validation
2	Calculator.java	Input Handling	Multiple operators allowed consecutively	Validation Issue	Medium	Improper input validation	Coding	Fixed	26-03-2026	27-03-2026	Added restriction
3	Calculator.java	UI Display	Result not cleared before new operation	UI Issue	Medium	Missing reset logic	Coding	Fixed	26-03-2026	27-03-2026	Fixed clear button
4	Calculator.java	Decimal Logic	Incorrect rounding for decimals	Calculation Error	Medium	Precision handling issue	Coding	Open	—	—	Needs improvement
5	Calculator.java	Button Events	Some buttons not responding occasionally	Functional Bug	High	Event listener issue	Coding	Fixed	26-03-2026	27-03-2026	Fixed event binding
6	Calculator.java	Negative Numbers	Negative input not handled properly	Logic Error	Low	Missing case handling	Coding	Open	—	—	To be fixed

Document Change History:

Ver.Rel. No	Release date	Modified By	Reviewed By	Approved By'	Remarks
v1.0	24-03-2026	Anshuman Chakravorty	Peer Reviewer	Faculty	Initial version of calculator application
v1.1	25-03-2026	Anshuman Chakravorty	Peer Reviewer	Faculty	Added input validation and fixed basic bugs
v1.2	26-03-2026	Anshuman Chakravorty	Peer Reviewer	Faculty	Improved UI and fixed division by zero issue
v1.3	27-03-2026	Anshuman Chakravorty	Peer Reviewer	Faculty	Enhanced error handling and optimized logic

Note :Since Testcases is baselined item, we need to keep history of changes. This sheet should be used to keep track of the changes.

PD-04-04
Ver 1.02 / 25-Jun-04

Testing Checklist

Project ID:	CALC-01	Project Name:	Java Calculator Application
Date:	26-Mar-26	Author(s):	Anshuman Chakravorty

Checklist details:

Sr. No.	Checklist	Yes / No / Not Applicable
Note:	This checklist should be used as a subset. Points that are applicable to your project should be taken from this checklist and a new checklist should be build that contains additional points specific to your project.	
Test Plan		
1	Was the testing procedure and guideline read/ understood by PL/ Team members ?	Yes
2	Are people identified for testing trained for testing activity?	Yes
3	Are test plan prepared?	Yes
4	Is the test plan consistent with the overall project plan?	Yes
5	Has a test schedule been explicitly defined?	Yes
6	Does test plan (PM-01-01) includes test preparation, test duration, and test reporting?	Yes
7	Are scheduling of test activities and allocation of resources done?	Yes
8	Are test resources and tools identified and recorded in PMP?	Yes
9	Does test plan address types of tests (Unit, Integration, System) and the sequence in which modules are integrated?	Yes
10	Are the test plans as defined in PMP followed?	Yes
Testing		
11	Is testing activity done by separate testing team?	No
12	Are the system level test cases / unit level test cases defined?	Yes
13	Are test cases prepared for all the functionality of the system?	Yes
14	Is updated requirements traceability table (PD-01-02) used during testing?	No
15	Has traceability to validation criteria/requirements been established? (Check for mapping of test cases with the requirements in RTT)	No
16	Does PL analyze the RTT and ensures that the test cases prepared provide sufficient test coverage?	Yes
17	Have test drivers and stubs been identified (for unit testing) and has work to develop them been scheduled?	No
18	Are test data provided by the client?	Not Applicable

19	Are test cases listed with their expected results?	Yes
20	Do the following test cases include test conditions for testing: - Security, mandatory fields, modal screens - Default values, validation of input - Error messages - Timing and performance	Yes
21	Is review plan for test cases identified ?	No
22	Are test cases reviewed for completeness and correctness?	Yes
23	Is the purpose of all different columns in test case understood correctly?	Yes
24	Does integration test cases cover all the interfaces (inter-modular and intra-modular) and application functionality?	Not Applicable
25	Is system testing performed in a simulated environment?	Yes
26	Does system testing consider interfacing to the external components, performance below expectations and failure under actual data volumes?	No
27	Are bugs logged into bug report(PD-04-02) for all types of testing?	Yes
28	Are bugs identified during testing categorized?	Yes
29	How is random testing planned and executed?	Yes
30	How is acceptance testing planned and executed?	Yes
31	Are combined testing test cases reviewed by both test team and customer?	No
32	Are defect summary and stage wise defect measurement data collected during testing phase?	Yes

B.3 Observations and learning:

During this experiment, the testing checklist was applied to the Java calculator application to evaluate the completeness and effectiveness of the testing process. It was observed that most of the core testing activities such as test planning, test case design, execution, and defect tracking were properly followed, ensuring the correctness of arithmetic operations and input handling. However, some advanced practices like requirements traceability, use of test drivers/stubs, and formal review processes were not fully implemented due to the small scale of the project. From this experiment, it was learned that a structured testing approach helps in identifying defects early, improving software quality, and ensuring reliability. It also highlighted the importance of proper documentation, validation techniques, and systematic testing methods even for simple applications like a calculator.

B.4 Conclusion:

In this experiment, a comprehensive testing checklist was applied to the Java calculator application to ensure the quality and reliability of the software. The process helped in systematically verifying all major functionalities, including arithmetic operations, input validation, and error handling. Most of the essential testing practices were successfully implemented, leading to a stable and functional application. Although some advanced testing activities like traceability and formal reviews were limited, the overall testing approach proved effective. This experiment demonstrates that even for small-scale applications, structured testing and proper documentation play a crucial role in delivering accurate and dependable software.